

Woolworths Online Order and Delivery System – Enterprise Systems Analysis, Design, and Modelling

Shawn Lee (23204035)

Table Of Content

Introduction	3
Organisational and System Analysis	3
Organisation Overview	3
Departmental Structure and Responsibilities	4
Problem Statement	6
Business Goals	6
Stakeholders and User Roles	6
Requirements	6
Requirement UML Modelling	7
Initial Structural Design and Implementation	7
Conclusion	7
References	8
Appendices	8

Introduction

- ☐ Brief intro to the project
- ☐ Purpose of the report
- ☐ Overview of what each section covers

Organisational and System Analysis

- ☒ ~~2.1 Organisation Overview~~
- ☒ ~~2.2 Departmental Structure and Responsibilities~~
- ☐ 2.3 Stakeholders and User Roles
- ☐ 2.4 System Scope, Boundaries, and Assumptions

Organisation Overview

Woolworths New Zealand (rebranded from Countdown) is a subsidiary of Woolworths Group Limited, an Australian publicly listed company. Woolworths New Zealand is a major supermarket chain whose headquarters are located in Mangere, Manukau, Auckland. Woolworth Group was founded by Percy Christmas on September 22, 1924, in Sydney, Australia. It is one of the 3 major supermarkets (Pak'n Save, Woolworths, New World) in New Zealand. Currently, there are 185 supermarkets^[1] in New Zealand and 20,000 employees^[1] working in New Zealand.

Woolworth's core values are providing convenience, value, range, and quality, and a motto of "The Fresh Food People"^[2]. Thus, Woolworths needs dependable information systems to manage its operations across shops, distribution centers, and digital platforms because it depends greatly on customer satisfaction. One of these operations that would be focused on in this report is the Online and Delivery System, since it offers greater convenience and the ability to "save time by ordering groceries online.

These advantages in these systems align with the organisation's values, in which the main focus is to provide the best experience for customers.

Departmental Structure and Responsibilities

In the ordering and delivery systems in Woolworths New Zealand, there are 4 critical departments in which forms the operation of the system. These systems are Store Operations, Supply Chain & Logistics, Customer Service, and IT.

- Store Operations

The store operations department for Woolworths New Zealand in 185 supermarkets^[1] focuses on restocking shelves and maintaining the food freshness of merchandise. In the context of an online ordering and delivery system, they are responsible for managing orders online by picking and packing them for delivery.

- Supply Chain & Logistics

The Supply Chain & Logistics department for Woolworths New Zealand focuses on managing stock movement from the supplier and distribution centres to the store. In the context of an online ordering and delivery system, they are responsible for assigning deliveries to complete orders and managing their delivery schedule, as 22%^[3] of orders online are fulfilled within 2 hours, and 57%^[3] of orders online are fulfilled within the same day of ordering, as well as ensuring there is enough stock for deliveries.

- Customer Service

The Customer Service department for Woolworths New Zealand focuses mainly on making sure that all customers are satisfied with the service and merchandise by handling customer inquiries, complaints, and issues. In the context of an online ordering and delivery system, this department is responsible for handling customer complaints and issues, assisting customers online, and processing refunds for incorrect or out-of-stock merchandise on the online ordering and delivery system.

- IT

The IT department for Woolworths New Zealand focuses on maintaining all digital platforms and technology systems, ensuring that user data is secure, developing improvements to the website, and resolving technical issues. In terms of the context of the online ordering and delivery system, the IT department is responsible for maintaining the system, integrating inventory, supply chain, and payment systems.

- People and Culture (HR)

The People and Culture Department for Woolworths New Zealand focuses on making sure that all employees are well-trained, supported in their well-being, and manage payroll and recruitment. In the 2025 annual report, mental health programs such as 'Sonder' and leadership skills are completed by over 55,000^[3] employees. This department also manages wage agreements and ensures there is an inclusive environment and culture.

- Finance

The finance department for Woolworths New Zealand is involved in managing the budget, managing the cash flow, planning, and analysing financial decisions, analysing financial risk, as well as producing financial reports. These responsibilities are important in overseeing the financial report of Woolworths New Zealand. Food's total sales reached \$8,286 million^[3] with normalised sales.

- Marketing

The marketing department for Woolworths New Zealand's main goal is to build brand awareness, build brand loyalty through rewards, and research the trends of the market, creating promotions and advertisements. This includes the marketing department managing the grocery shopping website, which contributed to having 2.1 million^[3] active everyday reward members in Q4 of 2025.

Problem Statement

- ☐ Creating a problem with the existing system could improve the system
- ☐ If there are no improvements that could be made, then you just recreate the existing system
- ☐ Need to improve on why the system is needed
- ☐ Need to describe both the customer and the business

As online shopping continues to grow in popularity due to its convenience, Woolworths New Zealand has identified a need to implement an online system of its own. Currently, Woolworths New Zealand does not have a dedicated online order and delivery system, relying on manual in-store shopping as the primary means of purchasing groceries, and has no system in which to manage the delivery of products when customers order online. Not having an online system also does not allow a notification to notify customers of delivery information, no platform to refund online or handle complaints, and limits marketing ability. This limits customer convenience and restricts Woolworths New Zealand's ability to meet growing demand for online shopping.

Furthermore, Woolworths New Zealand does not have an online or delivery system. From the customer's perspective on the manual purchasing of groceries, there is an efficient way in which customers are able to compare and decide different product choices based on their nutritional value, price, etc. Customers would also not be able to access Woolworths online, as there is no system that provides the ability for customers to purchase goods. The customer would not have the convenience or lack the physical ability to enter or check which store has a special promotion, opening, or closing date, or any news about the organisation. The functionality for the store operation employees is currently not able to visualize the total number of products offered in the store, thus needing to manually record and manage product stocks, which could increase the chances of human error. Employees would not be able to have statistical data on which product has the most demand, which leads to poor positioning of where products should be stocked.

The lack of an online system in which provide analytical data would make them rely on previous data about the stock levels and not current, real-time demand data, thus providing inaccurate data to the procurement specialist and the inventory controller, which could lead to inefficient distribution of supply to different store locations.

Woolworths New Zealand needs to develop an Online Order and Delivery System that allows customers to place an online order, in which customers are allowed to browse and search for products online. The customer should also be able to add products to their shopping cart and be allowed to check out. Customers would also be able to save a shopping list to reuse previous lists of orders to make it faster to shop for weekly used products. Customers will be able to create an account so that their details and saved shopping list are stored on the account for ease of access. The second part is the delivery system, which helps automate delivery order schedules so that delivery employees can be assigned to a schedule based on their location and availability to maximise the efficiency of delivery. The system should allow drivers to update deliveries in real time so that customers can view and be notified when their orders will arrive. Store operation staff would need to be able to receive information about the order and pick it ready for delivery. Shift managers need to be able to assign store staff by using the online system to delegate tasks as well as monitor the operations of tasks.

The system should improve customer convenience, streamline order fulfilment, and provide Woolworths New Zealand with a scalable platform for online grocery shopping.

Business Goals

User Requirements

Stakeholders and User Roles

- Customer (Create orders, Track the delivery)
- Supplier (Deliver stock,

- Store operations Employees (Pick merchandise, Pack merchandise)
- Delivery employee (Deliver order, schedule delivery)
- Manager (
- Customer Assistance
- IT Administrator
- Government & regulatory bodies

System scope, boundaries, and assumptions

- Assumptions
1. Assuming Woolworths has not implemented an order and delivery system and still relies on in-person grocery shopping.

Requirements

- ☐ Clearly defined functional requirements
- ☐ Clearly defined non-functional requirements
- ☐ Traceability between stakeholders and requirements

Requirement UML Modelling

- ☐ Use Case Diagram covering full system scope
- ☐ Minimum of five fully dressed Use Case Descriptions
- ☐ Correct use of include, extend, and generalisation relationships

Initial Structural Design and Implementation

- ☐ Initial UML Class Diagram
- ☐ Mapping between UML classes and planned C# classes
- ☐ Identification of anticipated design weaknesses prior to OOP application
- ☐ OOP-based implementation in C# that strictly conforms to the UML models.

Conclusion

References

1. Woolworths. (2024). Online Supermarket: Online Grocery Shopping & Free Recipes at [countdown.co.nz](https://www.woolworths.co.nz/info/about-us/our-company). Countdown NZ Ltd.
2. O'Connell, J., & O'Connell, J. (2024, November 19). 1987 The Fresh Food People. Australian Food Timeline. <https://australianfoodtimeline.com.au/fresh-food-people/>
3. WOOLWORTHS GROUP LIMITED. (2025). Annual Report 2025. <https://www.woolworthsgroup.com.au/content/dam/wwg/sustainability/reports/f25/Woolworths%20Group%20Annual%20Report%202025%20.pdf>

Appendices

☐ Github Link